

[L5] Chapters 9 & 10 - Dream machines and modes of computing

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- Big Blue was a sinister nickname because of the association with Big Brother

Role of Women

- There are generally very few women mentioned in the history of most sciences
- Science was for a long time claimed by man => their contributions were mostly underappreciated and sometimes mostly forgotten
- Esther Dyson was one of the most tech savvy people in the US during her time and wrote a newsletter that was read by most tech CEOs

Silicon Valley / European Valleys

- The place where the computer industry started boiling in the 70s

In Europe:

- There were no European valley-equivalents, because the valley was special
- European valleys were part of utopian city plans
- There were high tech companies and universities and there was interest for time-sharing within them
- There was not so much interest in the youth culture, which was crucial for the development of Silicon Valley
- Utopian city planning was part of planning culture in Europe in general
- Sophia-Antipolis is one of the more successful European valleys
 - South of France, near Nice
 - Beautiful scenery, near airport
 - It was built as a city where people would be born, live and work. It had housing, industry, recreation
- In general, this did not work as expected. People did not really want to live there. People would live there during the week and go home for the weekend, or even commute to the factories on a daily basis
- Twente was founded in 1964 and located right outside Enschede
 - People didn't want to move all the way to the east of the country as the west was the most densely populated
- In Europe there was interest for computers within the academic circles and high tech companies but not within the youth movement
- There were a few clubs founded:
 - Amateur Computer Club (ACC) by Mike Lord (Essex, UK, 1973)
 - Association Française des Amateurs Constructeurs d'Ordinateurs (AFACO) by J-C Ribes (Meudon, France, 1974)
- These groups were associated with the amateur computer enthusiast, but they didn't want to change the world, to create something new, who were appropriating with the idea of creating something new
 - They were appropriating computers, but were generally only interested in their immediate surroundings (their own club)
- The ACCN had a newsletter => people wanted to join together and learn from each other
- People would buy superfluous machines from local companies and started tinkering
- Unlike the US, the hippies in Europe did not blend with the computer enthusiasts, so it did not cause the huge enthusiasm that occurred in Silicon Valley

Appropriating the machine

- The Homebrew Computer Club (1975) (in the US) - people got together and wanted to show and discuss what they could do with the machine and how it could be useful to the rest of the world. This was not taking place in Europe
- Hobby Computer Club (HCC)
 - 1977, Netherlands
 - Resembled the homebrew computer club
 - Had activities for the members
 - Exchanged knowledge and software
 - Had a newsletter where programs would be featured
 - Had radio-broadcasting programs

- They had **NO** political agenda at all. They were the quintessential hobbyist
 - There was a sense of identity: if you came in, you had to know your stuff
- Other clubs:
 - PET Benelux Exchange (1979)
 - Philips Thuis Computergebruikers (1981)
 - Micro Computerclub Nederland (1984)
- Software would be swapped over mail in the form of tapes (many of them were games)

Squatter Movement:

- In the 80s, groups of people would squat in a home and create an anarchistic or communist community
 - Within this movement, the computer would be used, tinkered with and used for political purposes
 - There was a belief that surveillance was taking place through the machines by the government
 - Chaos Computer Club
 - Based in Hamburg, Germany
 - Trying to show to the people that computers and computer networks were antidemocratic and not always in their interest
 - They hacked Deutsche Bank, transferred money to their account and announced the bank the next day and returned the money
 - Once the media attention shifted, they were prosecuted for the hack
- Information Scientists were left out of computer science circles in the Netherlands
 - There were machines being used in the administration sectors
 - Before the 70s, there was a club, but not a hobby club: SSAA (Stichting Studiekring Administratieve Automatisering), founded 1958
 - Journal: *Informatie*
 - Providing information, courses and seminars
 - There was little overlap with the hobby computer club
 - They were culturally far removed. They were the people within the large companies with resources to buy computers
 - More "serious" users (used software like VisiCalc)
 - This appealed to large company managers but also to some hobbyists
- There were televised lessons about computers and microprocessors at Teleac
 - Chriet Titulaer was the director of the Dutch "Television Academy" and wanted to promote sciences. He moved on to the more popular topics (computers, and by extension, space flight)
 - This had an outreach far greater than the hobby computer club and the SSAA could provide

Computers in Education

In the Curriculum

- The computer was present in textbooks in the form of pictures
- Subjects that would show up are binary arithmetic, flowcharts and even programming (languages: Ecole (French branch of ALGOL), BASIC)
- It was not possible to have programming courses because there were no computers in the classes
- Some schools would have contracts with computer centers that would process students' programs, but it was only in major cities and very rarely even then
- It was hard to get pupils to think for themselves and learn programming
- Computers could help:
 - Helping the teacher teach
 - Adapting to the pupil's needs
- Students' tasks would be split into small steps that could be individually completed and checked
 - This was viewed to some extent like programming the child
 - Education could be viewed as a very technical issue
- The Skinner (1954) machine allowed the student to answer the question and only move on to the next one until the answer was right
- The concept of "Programmed Instruction" was also put into textbooks. This is an example of computing influencing education
- From the 80s, computers became common, with companies like IBM or Philips that would help schools build computer rooms